



Research Paper

Cluster randomized controlled trial of a phone-based caregiver support and parenting program for Syrian and Jordanian families with young children

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ABSTRACT

For refugee caregivers who may live in remote areas or be a highly mobile population, creating parenting programs that fit their needs and accommodate their mobility can be highly beneficial. In this article, we evaluate a 6-month, audio-only early childhood development (ECD) intervention delivered via phone (3 calls per month) to caregivers of Syrian and Jordanian backgrounds in Jordan. A sample of stipended community health volunteers (CHVs; $N = 99$) and their caseloads of caregivers ($n = 2,298$) was randomized to calls with health and nutritional content (control group) or calls with health, nutritional, and added parenting and caregiver-focused content (including content adapted from the Reach Up and Learn model). Hypothesized outcomes included reduced caregiver depressive and anxiety symptoms, parenting stress, and harsh discipline; increased parent-child learning activities and parenting self-efficacy. Exploratory outcomes (all caregiver-reported) included child development; and caregiver engagement in activities related to health and child learning following the calls. Exposure to the treatment condition resulted in a statistically significant reduction in caregivers' depressive symptoms ($d = -0.11$). No other statistically significant impacts were found. Implications of these findings for caregiver interventions are discussed.

1. Introduction

Parenting young children can be quite stressful, especially in the absence of key supports and resources or in the presence of substantial challenges and barriers. Given limited resources and/or stressors like forced displacement, caregivers may prioritize meeting children's basic needs, such as shelter, food, and clothing, over providing early learning and development opportunities (Yoshikawa et al., 2020). In addition,

parent's well-being, which has an impact on a child's development and well-being as well, may also be affected by such stressors (Eruyar et al., 2018; Miller et al., 2023).

1.1. The Role of Parenting Programs

Parenting programs have been utilized as a way to address both aims: provide parents with tools that can make them more successful and also

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improve parental well-being (Miller et al., 2023). Typically, parenting programs can be implemented through regular visitation to parents by a community worker (home visits) or offered as regular group sessions. Most impact evaluations associated with such programs show that, on average, they are quite effective in improving child developmental outcomes, but less is known on their impact on caregiver mental health (Jeong et al., 2021). A systematic review of 102 randomized controlled trials (RCTs) of parenting interventions across 33 low-, middle- and high-income countries (all focused on 0–3 year-old children) found that such interventions improve child cognitive, language, motor, and socio-emotional development; strengthen child attachment; and increase parent knowledge, positive parenting, and parent-child interactions (Jeong et al., 2021). Effects on cognitive, language, and motor development and parenting outcomes were larger in low- and middle-income countries (LMICs). Jeong et al. (2021) did not find any impacts on caregiver mental health (e.g. depressive symptoms), possibly because most parenting programs reviewed did not have an explicit focus on caregiver well-being.

Parenting programs may also play a role in mitigating the effects of conflict and displacement on child development. However, a systematic review of ECD program evaluations in conflict-affected areas yielded only three studies in LMICs (Murphy et al., 2018). The first, a group-based psychosocial intervention for mothers of young children in Bosnia, found positive effects on mother's mental health and child cognitive development. The second, conducted in Northern Uganda, found that psychosocial support for caregivers (mother / baby groups and individual home visits) improved maternal involvement and increased play materials. The third, an evaluation of group parenting sessions in Liberia, found that discussing the negative effects of violence on children strengthened caregivers' sense of identity as nurturers and protectors and decreased harsh discipline. Overall, these evaluations show positive effects on select (and differing) aspects of parenting, but few effects on child development and caregiver mental health.

1.1. The context: Reach up and learn and the Syrian refugee crisis in Jordan

One home-visiting-based parenting intervention that has been implemented (and evaluated) in a number of LMICs is Reach Up and Learn (RUL; Walker et al., 2018). RUL was created in 1986–7 based on the Jamaica Home Visit intervention. It provided one-hour weekly home visits for 2 years to families of children aged between 6 months and 2 years-old, at the beginning of the program. The RUL toolkit consisted of a training manual, structured-play curriculum that offers weekly or biweekly one-hour sessions between home visitors and caregivers that focuses on the child's play. Activities are sequenced and tailored to the child's developmental capacities, and are centered on three main components: activity demonstration, practice facilitation and feedback. The home visitor demonstrates a new parent-child activity with the child (often with a new toy or material or concept); the parent practices the new activity in the presence of the home visitor; and then the visitor provides immediate feedback.

The original RCT evaluating the Jamaica program showed significant positive impacts on child cognitive outcomes in the short run (Grantham-McGregor et al., 1991) and at ages 7–8 (Grantham-McGregor et al., 1997); 11 (Walker et al., 2000); 17 (Walker et al., 2005); 20 (Gertler et al., 2014); 22 (Walker et al., 2011); and 31 years-old (Walker et al., 2022).

In 2016, our implementing partner, the International Rescue Committee (IRC), in collaboration with Arab Resource Collective, a local not-for-profit, adapted RUL for Syrian and host community families in Lebanon, Jordan and Syria and began to integrate it into their programs in these countries (for an implementation report, see Wilton et al., 2021). From 2016 to early 2020, this implementation was part of a larger early childhood initiative in the region. Caregivers of children 6–42 months received weekly or bi-weekly in-person home visits over

6-months to 1-year, depending on the context. When COVID-19 halted in-person programming in March 2020, IRC's early childhood and development (ECD) team further adapted RUL in Jordan to allow for delivery through phone calls. Audio-only calls were chosen after consulting a sample of caregivers, who preferred audio-only calls as they worried about the data costs of sending/receiving videos and pictures or conducting video calls.

Despite the strong evidence on the effectiveness of parenting programs and RUL, there are some gaps that this study aims to fill. First, parenting programs in their current typical format (home visiting or in-person group sessions) may be difficult to implement in conflict-affected or rural contexts or with highly mobile, displaced populations. This suggests a need for parenting support programs that can effectively be delivered remotely, over the phone, for example (Ramachandran et al., 2022). This also suggests the need to evaluate the benefits and challenges of this unique delivery modality. However, to the best of our knowledge, no one has evaluated an audio-only, phone-based parenting program (the single published impact evaluation of an audio-only program focused on children diagnosed with diabetes in the US; Mackey et al., 2016). There are other parenting programs (most in response to COVID-19) that only use phone calls (e.g., Pashe Achhi, implemented by BRAC in Bangladesh), but they have not undergone impact evaluations (Ahmad et al., 2020). Most of the evaluated technology-based parenting programs use other media such as the internet (Baggett et al., 2010), have technology as an add-on to in-person programming (Carta et al., 2013), or have the phone call group as a low-dosage comparison group to higher intensity in-person programming (Guttentag et al., 2014). Such evaluations do not inform how feasible it is to replicate key interactions that make in-person parenting programs effective using audio-only phone calls. While phone-based parenting programs would have the benefit of reaching hard-to-reach populations and potentially parents who may be more comfortable with a phone call rather than an in-person interaction; delivering parenting programs remotely could compromise the quality of implementation, depending on the program structure. It may be difficult to replicate key interactions that make in-person parenting programs effective over purely audio phone calls. Demonstrating a new activity with the child, providing opportunities for caregivers to practice the activity, and providing real time feedback in-person, for example, are not possible on audio-only calls.

Second, very few studies have evaluated the impact of parenting programs delivered in Arabic-speaking countries. While there are many reasons to believe that successful programming in other contexts would transfer to Arabic-speaking countries, there are always concerns as to cultural mismatch (e.g., differing values, beliefs, socialization practices; Morelli et al., 2018; Weber et al., 2021). RUL, from which this study's intervention was adapted, has successfully been implemented in other contexts and improved child development (e.g., Colombia and Bangladesh: Attanasio et al., 2014; Hamadani et al., 2006), but has not been evaluated in Arabic-speaking countries. Current literature on Arab parenting does not offer a deep understanding of culturally specific parenting frameworks, preferences and beliefs in the first 1,000 days (Sim et al., 2018). Thus, it is unclear if assumptions embedded in the theories of change of parenting programs generally, and this parenting program specifically, hold in this context.

There are four recent examples of impact evaluations of parenting programs in Arabic-speaking contexts. All were in-person, parenting groups implemented in Lebanon, except one study that was implemented in Jordan as well (Lakkis et al., 2020). The first study, was a cluster randomized control trial study evaluating the effectiveness of a Preventive Parenting Program that took place over 7 weekly one-hour in-person sessions. The program had 3 main components: knowledge on infant/toddler psychological development; parenting skills including daily routines and play skills; and the participant's reflection on their representation of parenting and attachment. No significant effects were detected from this study (Dirani et al., 2022). The second study was an evaluation of the Mother-Child Education Program (MOCEP) in

Lebanon, found improved disciplinary practices and reduced parenting stress (Ponguta et al., 2020), but no effects on child development. It was conducted in three (non-refugee) communities and the program (originally developed in Turkey) focused on child development, importance of play, parenting practices, health/hygiene and maternal empowerment.

The third, also based in Lebanon but with Syrian parents of young children, tested the impact of the Caregiver Support Intervention: a 9-session program focused on psychosocial support and strengthening parenting through two aims: strengthening caregivers' parenting-related knowledge and skills; and improving mental health and well-being of caregivers (Miller et al., 2023). They found increased parenting quality; reduced psychological distress; and improved child psychosocial well-being (all caregiver-reported; Miller et al., 2023). The fourth study evaluated a pilot intervention that offered refugee parents 21 weekly 2-3 hours training sessions and discussions focused on ECD and mental health as well as parents' relationships with their children. The pilot study showed significant improvements in parents' mental health and wellbeing as well as a reduction in their perceived difficulties with their children (Lakkis et al., 2020).

Third, as mentioned before, impact evaluations associated with parenting programs have focused on child developmental outcomes but only occasionally assessed caregivers' mental health and well-being. Finally, while RUL has proven its effectiveness across varying contexts, no previous evaluation of a phone-based version of it exists, to the best of the authors' knowledge.

This paper reports the impacts of an audio-only version (i.e., regular phone calls with no Voice Over IP, text messages, video chats, or other mobile phone-based communication means) of an early stimulation and parent support program targeting Jordanian and Syrian caregivers living in Jordan. The program was evaluated through a cluster-randomized controlled trial (CRCT) to examine its effects on caregiver mental health, parenting, parenting stress, and parenting self-efficacy. Our preregistered hypotheses were that this program would reduce parenting stress while improving caregiver mental health and increasing parent-child activities, responsive parenting, and parenting self-efficacy and thus provide a model for future programs for hard-to-access populations. We also, as exploratory outcomes, examine whether this program affected caregiver-reported child development and engagement in activities related to ECD. The research was led by New York University's Global TIES for Children (NYU-TIES) in collaboration with the IRC and Sesame Workshop (SW) teams.

1.3. Research questions

This study is the first to evaluate a phone-based parenting program in the Middle East and the first RCT evaluation of an audio-only parenting program serving caregivers without diagnosed health problems or disabilities. Our research questions compare the focal intervention to receiving audio-only, phone-based health and nutrition support on its own:

- 1) What is the impact of a six-month, audio-only, phone-based adaptation of RUL in addition to health and nutrition messages (the program) on caregiver well-being (depressive and anxiety symptoms) and parenting (parenting stress, parenting self-efficacy, caregiver-child learning activities, caregiver responsiveness, and disciplinary practices)?
- 2) What is the impact of the program on child behavior and development?
- 3) What is the impact on reported activities / time use related to the RUL intervention?
- 4) Are program effects moderated by household nationality, child gender, or child age?

The first research question is preregistered with hypothesized positive impacts. The other three are exploratory. All outcomes are

caregiver-reported.

2. Methods

2.1. Participant characteristics

Families in our sample were 55 % Syrian, 44.5 % Jordanian, and 0.5 % from other nearby countries. The majority of the sample has lived in Jordan for at least several years. No participants reported living in refugee camps; just over a third (35 %) lived in Ramtha with the rest in Irbid. Caregivers (largely mothers) were 95 % female and 18-77 years old ($\mu=28.4$). The majority of caregivers (99 %) reported their age as 43 or younger and only three listed their age as greater than 50 (52, 60, 77), which aligns with the program's registration data indicating two were grandmothers.

Literacy rates were high: 88 % can read newspapers and magazines in Arabic, 6 % mostly can, and 2 % more-or-less can. Focal children (enrolled in intervention) were 49 % female and range from 1-57 months old ($\mu = 17$)⁴ with 98.5 % between 6 and 36 months old (the target age range of this study) when the program began two months later.

Most caregivers reported difficult financial situations: 11 % say (*often*) and 27 % (*sometimes*) not enough to eat. One in ten participants reported (*almost always*) restricting adults' food so children can eat; and another one in five (*more than half the time*). They reported their financial situations as worse than before COVID and notably worse than before the Syrian war. Caregivers also reported they (*almost never*) or (*less than half the time*) were able to: pay for basic needs (63 %) or heat their household (42 %); or had access to: internet (56 %), healthcare (45 %), cooking facilities (32 %) or clean drinking water (16 %); or had enough clothing (40 %).

Nine percent report having had a serious illness/injury in the preceding year. Just over half ranked their health as good (30 % *fair*, 13 % *poor*, and 4 % *very poor*). Caregivers reported that 7 % of focal children had a serious illness/injury in the past year and 3 % had poor or very poor health (with 81 % and 16 % in good and fair health, respectively). The average household had 6 people and 3.1 children and moved 1.3 times in the past 3 years. Participants reported low levels of available nearby support and low rates of seeking out others for help.

2.2. Sampling procedures

Community Health Volunteers (CHVs), hired by the IRC, recruited families with at least one 6-36 month old to receive health messages that were developed in response to COVID-19. Families were from the CHVs'

⁴ For 27 % of the 1,714 households in baseline and endline, a different date of birth (DOB) was given in each wave. For 174 of these, the month and year were the same and only the day differed. In these cases, we used endline DOB for all age calculations. For the remaining households, we looked at program registration data. If the child's month and year of birth were the same in one wave as they were in the registration data, we used that wave's DOB ($n = 216$; 113 matched baseline, 103 matched endline). If not, the families were called and re-asked the child DOB: 19 families re-gave baseline DOB, 24 re-gave endline DOB, 8 re-gave registration data DOB, and 5 gave a new DOB. We used the one given on the follow-up call in all these cases. Where the family was not reachable and all DOBs given across baseline, endline, and registration differed but were within 4 months of each other, we used the mean of them ($n = 4$). If the same month and same year (but not same combo) were given twice, then we used that month/year and most the common/mean day for DOB ($n = 5$). If none of these were true, we made child age/DOB missing and imputed it ($n = 8$). Realizing similar errors likely existed for those we only have baseline data for, we then compared baseline DOBs for those families ($n = 584$) to registration DOB. If month and year matched we kept as is. If within 4 months we used the mean ($n = 104$). If not within 4 months ($n = 74$) or no registration DOB on file anymore ($n = 2$) we dropped baseline DOB and imputed it.

social networks. Both the CHVs and their social networks were within our population of interest (Syrian refugees in Jordan and low-income Jordanian families). In the Jordanian governorate Irbid and the city of Ramtha, where this study took place, families started receiving health calls (focused on health messages based on the family's needs) in May 2020. In November 2020, families who met the inclusion criteria of having received at least 1 health call were offered a chance to participate in the study.

2.3. Sample size, power, and precision

Based on our power calculations for a two-level cluster randomized design with treatment at level two, the intended sample size was 2,450 participants. Assuming 20 % attrition (leaving a sample of 1,960 or approximately 20 families per each of the 99 CHVs); these power analyses (0.8 power; 0.15 intraclass correlation between CHVs; that individual baseline covariates, including baseline measures of our outcomes will explain 40 % of our individual outcomes; and that CHV covariates will explain 50 % of CHV outcomes (estimates drawn from Yousafzai et al., 2014), this goal yielded a minimum detectable effect size ($p=.05$) of 0.181 (as calculated using PowerUp!: Dong & Maynard, 2013). A total of 3,731 participants were assessed for eligibility which is having 1 child within 6 to 36 months. Of them 2,298 met the inclusion criteria (of having received at least 1 phone call) and agreed to participate. For both this final full sample ($n=2,298$) and our endline sample ($n=1,714$) our MDES remains 0.18: going down to 0.177 for the sample of 2,298 and up to 0.184 for the sample of 1,714. These changes in MDES are small because the treatment is delivered at the cluster level and thus the cluster size has a larger influence on power than the individual sample size.

Forty enumerators (24 female, 16 male) and 8 supervisors (5 female, 3 male) collected data over the phone at two time points: baseline (Fall 2020) and endline (Fall 2021). Participants were not compensated for their participation in the study. This study was approved for human subjects research by the institutional review boards of both the IRC and New York University.

2.4. Measures and covariates

This study had primary, exploratory and implementation outcome measures. The primary outcomes included: caregiver well-being, parenting, caregiver-child learning activities and disciplinary practices. The exploratory outcomes include the child's behavior and development. The implementation outcomes included caregiver report of activities related to the intervention. The majority of the measures, including sociodemographic and outcome measures, were first piloted in person in early 2019 in a pilot study of 171 households drawn from 36 CHVs during the in-person implementation of RUL. Revisions to wording of survey items for the local context were conducted. The measures were further piloted for usability over the phone in October 2020. Both baseline and endline data exist for all outcomes. The baseline psychometrics align with those at endline (presented here) and are available upon request. See Table 1 for a summary of all outcome measures used with sample items.

2.4.1. Caregiver well-being

We assessed caregiver well-being via self-reported depressive and anxiety symptoms.

2.4.1.1. Depressive symptoms. For this outcome, we used the 8-item Patient Health Questionnaire (PHQ-8). In the PHQ, respondents are asked a series of questions about frequency of depressive symptoms over the last two weeks. We replaced two weeks with one month to match the reporting period used for other survey questions. Both the PHQ-8 and the PHQ-9 (with an added question on suicidal ideation, which we

Table 1
Summary of measures

Outcome Measures	# items	Sample Item	α	Source/Notes
Caregiver Mental Health				
Depressive Symptoms: Patient Health Questionnaire (PHQ-8)	8	In past month, how often did you feel you have: little interest or pleasure in doing things? Options: Not at all, A few days, More than half of the days, Almost every day	0.82	(Kroenke et al., 2009)
Anxiety Symptoms: Generalized Anxiety Disorder (GAD-7)	7	In past month, how often did you experience: feeling nervous, anxious or on edge? Options: Not at all, Several days, More than half the days, Nearly every day	0.86	(Spitzer et al., 2006)
Parenting				
Parenting Stress: Parental Stress Scale (PSS)	8 (1 dropped)	Caring for [CHILD NAME (CN)] sometimes takes more time and energy than you have to give Options: Almost Never, Less than half the time, More than half the time, Almost Always	0.82	(Berry & Jones, 1995)
Parenting Self Efficacy: Nurturance, Discipline, Play, Routine	21 (5 dropped)	Managing [CHILD NAME (CN)] is difficult Options: Almost Never, Less than half the time, More than half the time, Almost Always	N/A	(Coleman & Karraker, 2003; van Rijen et al., 2014)
Caregiver-Child Learning Activities	5	Telling stories, reading picture books, naming / counting / drawing, singing, playing Options: Frequency in last week	N/A	UNICEF – MICS 6 (Kariger et al., 2012)
Positive and Negative Disciplinary Practices	10	Gave [CHILD NAME (CN)] something else to do to distract him/her (positive) Called [CHILD NAME (CN)] dumb, lazy, or another name (negative) Yes/No	N/A	UNICEF – MICS 6
Exploratory Child Outcomes				
Child Behavior - BITSEA: pro-social, externalizing, internalizing	33	Shows pleasure when succeeds (for example claps for self) Options: Not true	0.74 0.74 0.76	(Briggs-Gowan et al., 2004)

(continued on next page)

Table 1 (continued)

Outcome Measures	# items	Sample Item	α	Source/Notes
Child Development - ASQ-3: communication, gross motor, fine motor, problem solving	24 per form (6 per domain)	/ Rarely, Somewhat true / Sometimes, Very true / Often, N/A 22 mos., comm.: Without you giving clues, can [CHILD NAME (CN)] carry out 3 of these kinds of directions? A) "Put toy on table." B) "Close the Door." C) "Bring me a towel." D) "Find your coat." E) "Take my hand." F) "Get your book." Yes, Sometimes, Not yet	0.71 0.73 0.74 0.61	(Squires & Bricker, 2009)
Exploratory Implementation Outcomes				
CHV encourages / freq. of health / ECD activities	6 (ran as single items)	Can you tell us how many days you spent on those educational activities? Options: vary by question	N/A	

omitted) have been translated and administered with Syrians in Greece (Poole et al., 2018) and Jordan (Jones et al., 2022). Item performance was first assessed via confirmatory factor analysis (CFA) using M Plus in which depressive and anxiety symptoms were assessed simultaneously. The model had good fit and depressive and anxiety symptoms formed separate, but highly correlated ($\alpha = 0.74$), subscales as anticipated. Factor loadings were similar enough to suggest mean scores could be used ($\alpha = 0.82$).

2.4.1.2. Anxiety symptoms. The Generalized Anxiety Disorder (GAD-7) questionnaire includes 7 items tapping feelings of nervousness, not being in control of worrying, and trouble relaxing. As with the PHQ-8, we replaced past two weeks with one month. The GAD-7 was previously administered to Syrian refugees in Lebanon (Doumit et al., 2020) and Jordan (Jones et al., 2022). Item performance was assessed alongside depressive symptoms (see above: $\alpha = 0.86$).

2.4.2. Parenting

For parenting, we measured feelings about parenting (parenting stress and parenting self-efficacy) in addition to parenting behaviors using both at-home and scenario-based parenting behaviors.

2.4.2.1. Parental stress. We translated and piloted the 18-item Parental Stress Scale (Berry & Jones, 1995) in Modern Standard Arabic (MSA) in 2019. Based on this, we re-translated the scale into Levantine dialect; dropped the 9 positive items (which had little variance and gave no additional information about parenting stress); replaced the *strongly disagree-strongly agree* Likert scale to reflect frequency (*almost never, less than half of the time, more than half of the time, almost always*); and made the language more specific (e.g., referring to the focal child directly by name). The remaining 9-item scale performed better without the second item ('You worry about whether you are doing enough for (focal child)'). Factor loadings were similar enough to support using mean scores for the remaining 8 items ($\alpha = 0.82$).

2.4.2.2. Parenting self-efficacy. Parenting self-efficacy (PSE) denotes

caregivers' belief in their ability to be good caregivers and provide the right resources and environment for their child(ren) (Albanese et al., 2019). We used the Self-Efficacy for Parenting Tasks Index-Toddler Scale (SEPTI-TS) short form, which includes 26 items across four domains: nurturance, discipline, play, and routine (van Rijen et al., 2014). While the long form, 7-domain SEPTI-TS has been used in two prior studies with Palestinians (Fass et al., 2018; Khoury-Kassabri et al., 2014), as far as we know, neither form has been used in Jordan or with Syrian refugees. The SEPTI-TS is scored based on level of agreement with statements such as "I am able to get actively involved in playing with my child." Negative statements are reverse-coded. As with parenting stress, we changed the answer options to reflect the frequency of caregivers' experiences of those feelings.

In our context, five items (one discipline, two play, and two routine) did not perform well. The model fit best with these five items dropped. Factor loadings for nurturance and discipline were disparate enough (e.g., ranging from 0.47 to 0.71 for discipline) to suggest it would be better to use factor and not mean scores in order to allow items to have different weights on the overall subscale score. For consistency, we used factor scores for all four PSE subscales (model / factor loadings available on request).

2.4.3. Caregiver-child learning activities

The Family Care Indicators measure (Kariger et al., 2012) assesses the number of days per week the caregiver or another adult does each of 11 early learning activities with the focal child. Exploratory factor analyses yielded two subscales in our data: traditional school readiness activities (reading, telling stories, naming/counting/drawing) and singing/playing. The other 6 activities (e.g. items that may apply more to older children such as helping with chores) did not load on any factor. A CFA with these two factors showed good fit with a correlation of 0.547 between factors. Given disparate factor loadings within factor, we use factor, and not mean, scores for both subscales (model / factor loadings available on request).

2.4.4. Disciplinary practices

We used the child discipline subscale from UNICEF's Multiple Indicator Cluster Survey-6 (MICS-6; UNICEF, 2020) for children under 5 years. The original MICS-6 asks caregivers to report (yes) or (no) to the set of 10 items. To allow more variability, we asked caregivers to recall whether, over the past month, they had engaged in each behavior (*never*), (*once a week*), or (*more than once a week*). CFA revealed good model fit for a 2-factor model of positive (3 items: e.g., forbade something they liked) and negative (7 items: e.g., shouted, yelled, or screamed at) discipline practices. Given variability in factor loadings, factor scores were used in the data analyses (model / factor loadings available on request).

2.4.5. Exploratory child outcomes

2.4.5.1. Child behavior. To measure child behavior, we adapted and translated the Brief Infant-Toddler Social-Emotional Assessment (BIT-SEA; Briggs-Gowan et al., 2004). First, we dropped 'the child will not touch an object because of how the object feels' due to difficulty in translating it. Second, we dropped all severe behavior problem items due to a lack of referral services. This left 33 (of 42) items. Third, we added examples to items identified by caregivers during piloting as being confusing. Fourth, we added a 'not applicable' option at endline based on caregiver feedback from baseline.

For analyses, questions where N/A made sense and was selected (e.g., asking about child behavior on being left which does not apply if child has not been left) were re-coded as missing and questions where N/A did not make sense (e.g., asking if worrying a lot is a serious issue, not applicable here is the same as saying that it is not a serious issue) were recoded as zeroes. Analyses were re-run with all N/As coded as missing

as a robustness check.

We ran psychometrics on this adapted measure rather than use traditional BITSEA scoring. Through factor analyses we found that five items did not perform well. The remaining 28 loaded onto three subscales: pro-social, internalizing, and externalizing behaviors. The pro-social subscale aligns with BITSEA's competence subscale nearly exactly. Remaining items from BITSEA's problem behavior subscale split into internalizing (e.g., 'worries a lot or is very serious') and externalizing (e.g., 'is destructive: breaks or ruins things on purpose'). Factor loadings were similar across items in each subscale; therefore, mean scores were used for analytic models (pro-social: $\alpha = 0.74$; externalizing: $\alpha = 0.74$; and internalizing: $\alpha = 0.76$).

2.4.5.2. Child development. We measured child development using the Ages and Stages Questionnaire (ASQ): a caregiver-reported measure covering communication, gross motor, fine motor, problem solving and personal-social skills (6 items each). The ASQ has different forms for specific age groups between 0 and 66 months (Squires & Bricker, 2009). Caregivers are asked if their child can perform various actions and answers are totaled by domain. The ASQ has been validated in Lebanon (Charafeddine et al., 2013) and is available from the publisher in Modern Standard Arabic. For our study, we removed the personal-social domain because the questions overlapped with BITSEA (above). Following ASQ scoring guidelines, if 1–2 items (out of 6) were skipped we used the average of the answered items and if more than 2 were skipped we scored that domain as missing. We also dropped ASQ scores for any child given a form 4+ months off their age after our date of birth (DOB) adjustments.

2.4.6. Implementation outcomes

Lastly, we explored impacts on caregiver report of activities related to the calls themselves. At endline, we asked caregivers six additional questions: i) 'On the calls, did the CHV encourage you to do the following health activities, such as healthy eating, exercising, breastfeeding visiting doctors, family planning or getting vaccines for children?'; ii) 'If yes, after the call, can you tell us how many days you spent on those health activities?'; iii) 'How many hours did you spend on those health and nutrition activity tasks on average per day?'; iv) 'On the calls, did the CHV encourage you to do educational activities, such as read, play games, or sing songs with your child?'; v) 'If yes, after the call, can you tell us how many days you spent on those educational activities?'; vi) 'How many hours did you spend on those educational or play tasks on average per day?'. Each was tested as a single-item outcome.

2.5. Data collection

A caregiver survey, including the abovementioned scales, was programmed using Open Data Kit (ODK) language to facilitate data collection using Kobo Collect Form⁵ via tablets (Poloju et al., 2022). Enumerators were trained on using the tablet to administer the survey to participants. After CHVs recruited, registered families to the program and delivered at least 1 health call, enumerators called these families, read the consent form and offered them a chance to participate. Baseline data collection took place from November 2020 to February 2021. Endline data collection took place between November 2021 and January 2022. Baseline and endline were almost one year apart as we were delayed in obtaining approvals for endline data collection.

The average time to complete the baseline survey (including reading out the consent form) was 80 minutes; families were therefore offered to complete it in one or two calls. For those opting for two calls, the first call included core questions in addition to a set of questions. The

additional set of questions was randomized to call one or call two to better distribute missing data if the second call was not possible. The questions were written in colloquial Arabic (not Modern Standard Arabic) to fit the conversational nature of the survey where enumerators read the questions and choice options to respondents and recorded the answers on the Kobo form. We shortened the endline survey (around 45–50 minutes; also including obtaining verbal consent) by removing all time-invariant questions (e.g. household inventory of family members, location, socio-economic segment scales; SES). As with baseline, participants were also given the option to complete it in one or two calls.

During their 5-day training on the caregiver survey (including data collection logistics, and research ethics), enumerators helped refine data collection tools and advised on the wording of survey questions. Enumerators underwent an additional 3-day refresher training prior to endline data collection.

Of the 2,298 baseline families, 1,714 families (74.6 %) participated in endline (see Fig. 1 for a complete CONSORT diagram; see Attrition Rates in Supplementary Material). There is no evidence of differential attrition from the study by treatment assignment (assessed by individual *t*-tests followed by a joint test of significance of differences). There is evidence that, in both treatment and control, families with higher baseline SES (i.e., education, income, resources), Jordanian families, and those who dropped out of the program were less likely to participate in endline. While we use the full sample for all analyses (imputing missing values), these differences suggest possible external validity concerns.

2.6. Intervention

The program evaluated in this study was implemented in the governorate of Irbid and city of Ramtha in Jordan from March to September of 2021. A two-arm CRCT was conducted, with both groups offered 3 audio-only phone calls per month for 6 months. The control group received health calls only while the treatment group received additional ECD content. Ninety-nine stipended, full-time Syrian and Jordanian CHVs implemented the program, some of them were offering the service in-person prior to switching to the phone modality. The average caseload for each CHV during the intervention was 31 families.

The program encouraged caregivers to practice specific developmentally supportive activities with children using positive parenting discipline practices. The intervention also aimed to build collaborative relationships between CHVs and caregivers. The theory of change hypothesized that supporting caregivers to improve and increase interactions with children, and to better understand the importance of ECD and how they can foster child development, will increase caregivers' self-efficacy and decrease parenting stress, thereby decreasing caregivers' overall anxiety and depressive symptoms. The improvements in caregiver well-being and the improved caregiving reinforce each other, as improved caregiver well-being helps caregivers to have more positive interactions with children, and more positive interactions with children in turn improve caregiver well-being. In the longer-term, this theory of change posited that these intermediate outcomes ultimately improve children's development across a range of domains – social-emotional, cognitive, and physical.

The RUL ECD content included information on child development and why it is important for caregivers to support children's development, as well as guidance for caregivers on how to perform specific activities with children to support different aspects of development. The RUL activities are designed to develop children's language skills, intellectual development, emotional and social behavior. RUL activities were divided according to the age groups of children, with call content targeted to month and week of child age. For example, a caregiver of a child who was 2–4 months old received content targeted to that age group. Within the 2–4-month-old age group, there are 9 scripts for each week in that age group. So, a child who is 2 months and 3 weeks old received a call specifically for that week in the age group, while a child who is 4

⁵ KoboCollect is an android-based data collection app that is particularly beneficial for remote areas and areas where internet connectivity may not be consistent.

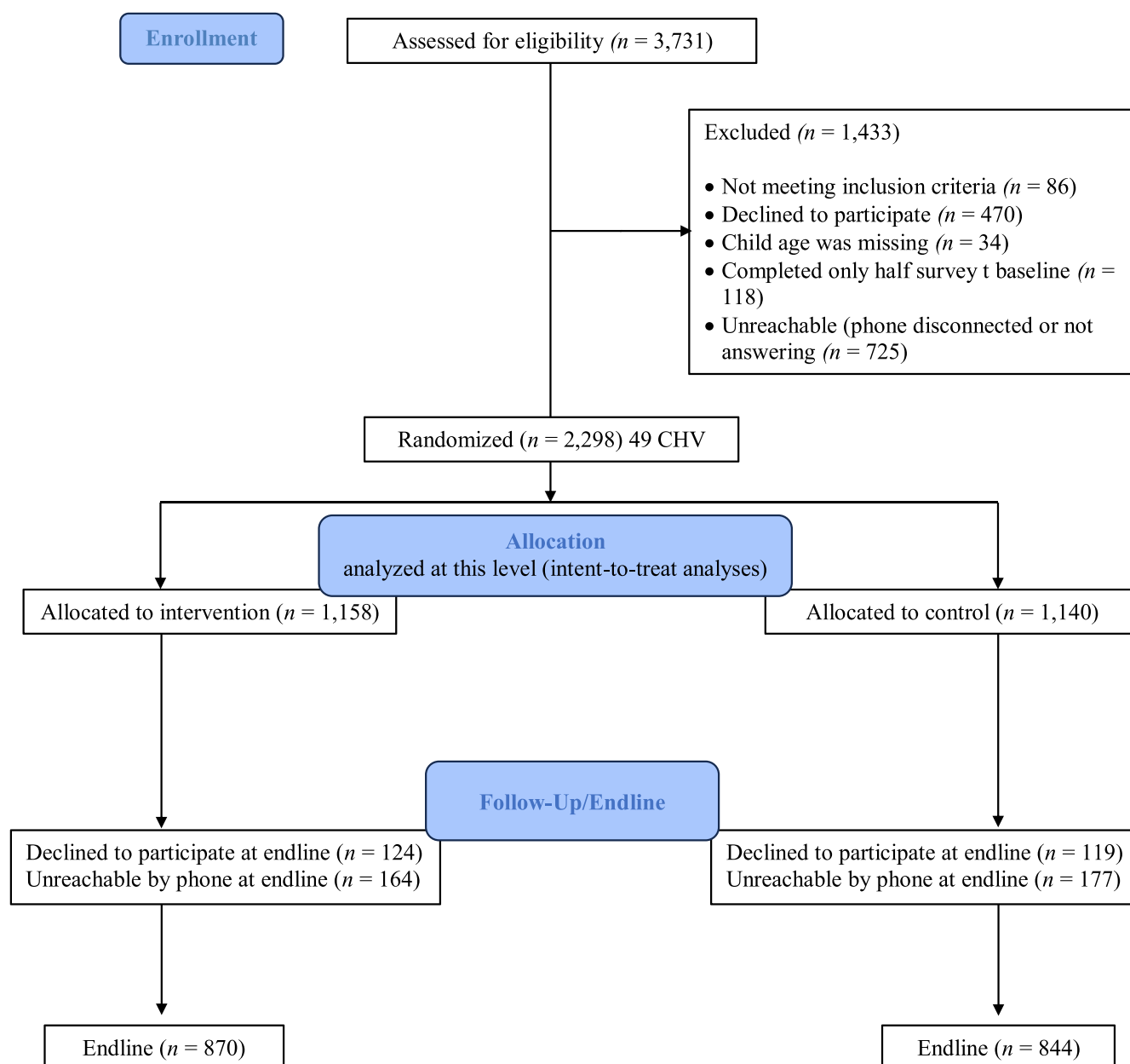


Fig. 1. CONSORT flowchart of participants

months and 2 weeks old received a different call with content tailored to that specific age. Call content exists for children ages 6 to 42 months, divided into three-month age groups. In general, the program aims to provide caregivers with knowledge related to the development of their children and to improve the methods of interaction with children in terms of communication, play, and learning through various activities, games, and songs that caregivers can implement using basic household items, such as plastic bottles, bottle caps, and cups.

The health and nutrition content and the RUL content are distinct, but related, as nutrition and physical health are important aspects of children's overall growth and development, which is why the IRC team thought it made sense to integrate health and ECD services. Some of the health and nutrition messages were targeted to the same age group of children as RUL – for example, feeding practices, nutrition, and vaccination. Health messages focused on children's nutrition and physical health, while RUL messages focused on caregiver-child activities that support cognitive and socioemotional development. The health and nutrition messages did not include content on the importance of

supporting children's development through interaction and play, but these topics did occasionally come up in CHV conversation around the health messages in an informal way. Importantly, the RUL messages focused on specific activities caregivers could do with children to support their development, and these types of activities were absent from the health messages.

Both the control and treatment groups received content focused on health and nutrition covering a range of topics, with some offered to all families (e.g., personal hygiene, food safety, COVID-19 precautions) and some offered based on characteristics and health needs of household members (e.g., the importance of and where to go for postnatal care, nutrition and immunization messaging targeted to child age).

The treatment group received additional RUL content including information on child development, caregivers' roles in child development, and guidance on how to do specific activities with their child(ren) to best support different aspects of development (i.e., language / cognitive skills, social-emotional behavior, self-confidence). RUL activities were grouped according to age (e.g., 2-3, 4-5, 6-7 months old) and content

focused on activities readily implementable using basic household items.

Control calls were intended to last about 18–23 minutes (3-minute greeting / check-in; 10–15 minutes on health topics; 5 minutes to summarize call content and address caregiver questions), with the additional content in the treatment group adding 7–10 minutes per call. Costs of these calls were covered by the IRC. While the health and nutrition messages did not include content on supporting child development through interaction and play (focusing instead on nutrition and physical health), these topics did occasionally come up informally in CHV conversations around the health messages.

2.7. Random assignment

CHVs, enumerators and participants were all blinded to the randomized design of the study. While the intervention was delivered individually to caregivers, randomization happened at the CHV level. After baseline data collection, 98 CHVs were matched based on a vector score into 49 pairs and randomized, along with the families they serve, to treatment or control within each pair (the 99th CHV was randomized separately).

We generated the pairs through an iterative process of data reduction using Principal Component Analysis (PCA) that maximized variance coverage. This resulted in a vector score (via which CHVs were matched) composed of: CHV caseload (e.g., caseload size, children's and caregivers' ages / nationality, caregiver education, number of health calls already received by families in caseload); and CHV's individual characteristics (e.g., their gender, age, nationality, experience, education). Paired matching was done to improve baseline balance given treatment assignment at the CHV level and a sample of only 99 CHVs.

The IRC kept a roster of trained CHVs for each condition to quickly replace CHVs who left the program. Control CHVs were trained to deliver health-only calls while treatment CHVs were trained to deliver health and RUL calls. If, for example, a CHV left⁶ who was in the control group, they were replaced with another trained control CHV. This occurred to 29 % of the CHV pairs. However, CHV demographics were not significantly different between initial CHVs and those that replaced them. Caseloads remained the same across CHV transitions.

2.8. Treatment fidelity

The IRC team conducted a content analysis study that indicated the treatment calls were in fact 5–7 minutes longer and included ECD content. The intervention was implemented as intended in terms of the structure and content of calls and adherence to treatment assignment. Actual dosage, however, was lower than intended, both in relation to the number and the length of calls. On average families received 12.3 calls (out of an intended 18) over the course of the study (13.2 for control and 11.4 for treatment). Evidence from a subsample of recorded calls suggests that the calls were also shorter than intended. Control-group calls lasted 20.6 minutes on average (instead of the intended 18–23 minutes), while the treatment-group calls lasted 26.0 minutes (instead of the intended 25–33 minutes). Time spent on health and nutrition topics within calls was found to be comparable across groups. The RUL/ECD content was only used in the treatment (as intended).

Among the 2,298 caregivers, 228 treatment and 182 control families dropped out after baseline but before the program started. Some caregivers (67 treatment, 232 control) requested and received additional (more than 18) calls. Most (75 %) of these families received only one or two additional calls.

In addition, CHVs delivering the intervention were thoroughly trained and consistently supervised to ensure standard implementation

of the intervention. All CHVs (in groups of 25 at a time) received 5 days of training for 6 hours per day via Zoom on the health and nutrition content. The treatment group CHVs received an additional 5 days of training on ECD and RUL content. This training included role-playing practice calls as well as additional training on active listening, initial greetings (including a well-being check in), responding to caregiver inputs and questions, stress management tips, and checking-in on (and providing feedback relative to) how caregivers used or tried to use activities from the last call.

CHVs also received ongoing supervision, including supervisors listening to CHV calls once a month (with caregiver consent) and scoring CHV performance during the call based on a structured observation tool. After these calls, supervisors coached CHVs to improve performance. CHVs also had monthly Zoom group meetings, separate for treatment and control CHVs, so that managers could brief CHVs on program logistics and answer questions.

Only families who were already enrolled in the program and had already received at least one phone call with the health-only content during the baseline from November 2020 to February 2021 were included in this study. If families dropped out during the intervention, they were still included in the study as we followed an intent-to-treat model. If CHVs replaced these families with new ones, the new families were not included in this study.

2.9. Analytic Approach

There was substantial (19 %) CHV turnover during this study. While CHV characteristics are still well balanced across pairs at endline, the length of CHVs' working with their families is not balanced; as the matched pairs did not experience the same attrition at the same time. This raises concerns about estimating treatment effects within matched pairs. Given this, we shifted from a primary analytic model based on the matched pair design to a to linear/logistic mixed model (depending on if the outcome is continuous or binary), in which households are nested within CHV (such that any consequences of CHV turnover, timing of turnover, etc. is shared variance within each group). We still retain the benefit of the original matched design to the degree that it improved our randomization and baseline balance. We also run robustness checks using within-pair fixed effects in a model otherwise similar to the below and within-pair fixed effects in a differences-in-differences model with time nested within household. All models yield very similar coefficients.

Our primary analytic model is:

$$y_{ij} = \gamma_0 + \gamma_1 \ell_{x_j} + \gamma_{2-26} \mathcal{X}_{s_{ij}} + u_j + \epsilon_{ij}$$

Where y_{ij} is the outcome variable being tested for household i assigned to CHV j ,

γ_0 is the mean outcome level for control CHVs,

γ_1 is the main treatment effect (mean difference in outcomes between treatment and control),

γ_{2-26} are the coefficients on the 24 covariates (plus the baseline measure of y_{ij})

u_j is the random error associated with CHV, and

ϵ_{ij} is the random error associated with household assigned to CHV j

The 26 covariates included are: caregiver gender (95 % female), child gender (49 % female), caregiver age (years: $\mu=28.43$), child age (days old at the time the outcome was assessed which varies by whether the outcome was asked on the first or second endline phone call: $\mu=874.67/876.39$ or 28/29 months old), if household is Jordanian ($\mu=45$ %, with rest nearly all Syrian), caregiver ($\mu=3.48$)/child ($\mu=3.79$) general health (single-item questions: *very poor* (1) to *good* (4)), if caregiver (9 %)/child (6 %) had a serious injury in past year (each yes/no), caregiver mood/general satisfaction (*very poor* (1) to *good* (4): $\mu=3.27$), caregiver disability (sum of 0 (*no difficulty*) to 3

⁶ CHVs left their position usually for better job opportunities or to move to another country (some CHVs were Syrians).

(cannot at all) for six questions: seeing, hearing, walking, memory, self-care, and communicating; $\mu=1.31$), eight indicators of caregiver social support (number of friends ($\mu=0.85$) /family ($\mu=1.60$) nearby that can rely on, have someone (yes/no) can go to for help with child (82 %)/ resource (82 %)/ personal problems (85 %), and frequency (from *never* (0) to *more than once a week* (5)) of seeking help for child ($\mu=2.02$) / resource ($\mu=2.10$) / personal ($\mu=1.28$) problems, a household resources factor score (financial situation, perceived wealth, food security, ability to buy/access to basic needs such as clean water, heat, healthcare, internet), and an educational SES factor score (own, spouse's, mother's, and father's highest education level), caregiver literacy (from 0=*cannot read food/drink labels* to 4=*can absolutely read newspapers/magazines*: $\mu=3.76$), location (Ramtha 35 %, Irbid 65 %), number of total people ($\mu=5.96$) / and children ($\mu=3.19$) in household, and number of residential moves in the past 3 years ($\mu=1.33$). We also control (when able) for the baseline measure of the outcome being ran.

We grouped the outcome variables into conceptual groupings following a previous exemplar (see Benjamini & Yekutieli, 2001; Mag-nuson et al., 2022; Noble et al., 2021; Schochet, 2008) and conducted *p*-value adjustments using false discovery rate methods (Benjamini & Yekutieli, 2001), to adjust for multiple hypothesis testing. Effect sizes are calculated by dividing the η^2_g estimate by the pooled standard deviation of the outcome.

3. Results

3.1. Missing data

Some data was missing due to: attrition; non-completion of second survey call and a version error with the Kobo form that led to some families ($n=299$, 17 %) not being asked to report on activities and time use. There was also a small amount of item level missingness (skipping questions) at baseline and endline.

For variables like father's education and parenting self-efficacy, that are part of a larger factor score, this missingness is accounted for using full information maximum likelihood within the CFA itself. For all other variables, missing data was imputed using MI chained in Stata 15 to create 100 complete, imputed datasets on which all analyses were conducted. All data were imputed in one model, but endline and treatment assignment variables were omitted from runs predicting baseline variables to not allow treatment assignment to influence pre-treatment scores.

Analyses are presented across the full intent-to-treat sample ($n=2,298$) with endline data imputed for those only in baseline. We report the flow of participants in Fig. 1. While we know that those who did not participate at endline look different on average than those who did (higher SES, more likely to be Jordanian families were more likely to have left the program), we still have substantial overlap (e.g., higher SES families who did participate in endline) from which to impute these data. We also ran analyses on only the participants who participated in endline ($n=1,714$) and got nearly identical results with the same substantive takeaways.

3.2. Impact on primary outcomes

We report the following outcomes for both Syrian and Jordanian participants. For our primary pre-registered outcomes (caregiver well-being and parenting; Research Question 1), the majority of ICCs at the cluster level were below .001; the highest were playing/singing at .02; anxiety symptoms; and learning activities at .03. We found a statistically significant, small reduction in caregiver depressive symptoms (see Table 2). This result was robust to *p*-value adjustment for multiple hypothesis testing ($n=2298$, $\beta=-0.06$, $p=0.021$, adj. $p=0.044$, effect size (ES) = -0.11). However, we found no statistically significant impacts on caregiver anxiety, any of the parenting outcomes, or parenting self-efficacy.

3.3. Impact on exploratory and implementation outcomes

For our exploratory outcomes of child behavior and development (Research Question 2), ICCs at the cluster level were below .001. We found no statistically significant effects on the BITSEA or ASQ measures. As for our implementation outcomes (Research Question 3), we did not find statistically significant impacts on report of CHV encouragement of and time spent engaged in key health/nutrition and ECD activities. Finally, we examined moderation effects (Research Question 4) starting with our three preregistered moderators (household nationality, child gender, child age). We found no moderation effects for any of these. We also explored baseline social support and baseline level of depression as potential moderators for depressive symptoms but find no evidence for either. Indeed, none of the effects on any outcome significantly vary by baseline level of that outcome.

4. Discussion

This study is the first experimental impact evaluation of an audio-only parenting and caregiver intervention in a low- or middle-income country, as well as the first in a humanitarian context. The program, an adaptation of the Reach Up and Learn (RUL) model developed in Jamaica (Walker et al., 2011), was implemented as part of Ahlan Simsim, a larger ECD initiative providing caregiver and child-focused ECD programs and mass media to Syrian refugee and host community families in Jordan, Syria, Lebanon, and Iraq (Wilton et al., 2021).

For one of our primary outcomes, caregiver well-being (defined as depression and anxiety), we found a statistically significant, small reduction in caregiver depressive symptoms. This finding is in line with a meta-analysis that examined 18 RUL impact evaluation studies and found an average decrease in maternal depressive symptoms that was not, however, statistically significant (Jervis et al., 2023). We did not find evidence of reductions in anxiety symptoms or parenting stress, nor increases in parent self-efficacy. This is a different finding from what Lakkis et al. (2020) found where parenting stress was reduced after 21 three-hour in-person sessions for Syrian parents in Lebanon. However, these sessions offered a more intensive dosage than our RUL dosage and it may be that the sources of anxiety may not have been as malleable in a relatively light-touch phone-based intervention as depressive symptoms. Anxiety is a distinct, yet highly correlated, mental health phenomenon from depression that has also shown increases during the COVID-19 pandemic. It is also the case that treatments for anxiety typically use a different set of techniques than those for depression, meaning that what effects depression symptoms might not have the same effect on anxiety. Overall, our results indicate that program impacts did not differ by any examined population characteristics.

Despite the emphasis on parent-child activities in the intended intervention, we did not find impacts on any of our parenting outcomes or on parent-reported child behavior and development. It may be that the dosage of the intervention was too low in looking at these outcomes. The original RUL program provided weekly in-home visits of 60 minutes each for 2 years. Adaptations since have provided 1-year versions of RUL at scale with positive impact on cognitive development (e.g., the national impact evaluation of Cuna Mas in Peru; Araujo et al., 2021). The current adaptation provided calls three times a month for 6 months, with ECD content limited to roughly 7-10 minutes per call. This is substantially less contact time than any previously evaluated in-person version of the program: approximately 126-180 minutes of intervention (this study) as compared with previously studied in-person versions of RUL that included between 3,000 (1-year program) and 6,000 minutes (2-year program).

As mentioned before, RUL depends on demonstration, practice and feedback. All three of these components, and the ability to tailor the component to the child's level of development, are compromised in an audio-only intervention. In contrast, mobile phone-based parenting programs that incorporate video have shown impacts for both parenting

Table 2

ITT effect estimates with effect sizes and *p*-value adjustments for multiple hypothesis testing within variable groupings (groupings 1 and 2 are preregistered; groupings 3 and 4 are exploratory) for full sample (*n* = 2,298)

Variable Grouping	Outcome	Hypoth. Direction	N	Health Only μ (pooled SD) (<i>n</i> = 1140)	ECD + Health μ (pooled SD) (<i>n</i> = 1158)	ITT Estimate	Effect Size	95 % CI	<i>p</i> -value	adj. <i>p</i> -value ^a
1	Depressive Symptoms (PHQ-8) (<i>mean score; range 0-3</i>) ^b	-	2298 ^c	0.99 (0.61)	0.93 (0.56)	-0.06*	-0.11*	[-0.20, -0.02]	0.021	0.044
1	Anxiety Symptoms (GAD-7) (<i>mean score; range 0-3</i>)	-	2298 ^c	1.23 (0.72)	1.23 (0.71)	-0.02	-0.03	[-0.27, 0.43]	0.489	0.324
2	Parenting Stress (PSS) (<i>mean score; range 0-3</i>)	-	2298	1.26 (0.67)	1.21 (0.64)	-0.04	-0.07	[-0.16, 0.03]	0.173	1
2	Parenting Self Efficacy (PSE): Nurturance	+	2298	-0.09 (0.49)	-0.09 (0.53)	-0.00	-0.01	[-0.11, 0.09]	0.847	1
2	Parenting Self Efficacy (PSE): Discipline	+	2298 ^c	-0.04 (0.52)	0.01 (0.53)	0.04	0.08	[-0.02, 0.17]	0.104	1
2	Parenting Self Efficacy (PSE): Play	+	2298	-0.02 (0.62)	-0.02 (0.67)	-0.01	-0.01	[-0.10, 0.09]	0.863	1
2	Parenting Self Efficacy (PSE): Routine	+	2298	-0.01 (0.63)	-0.02 (0.66)	-0.01	-0.02	[-0.11, 0.08]	0.715	1
2	Traditional parent-child learning activities (FCI)	+	2298 ^d	0.01 (1.07)	0.01 (1.09)	0.03	0.03	[-0.06, 0.12]	0.542	1
2	Singing/playing with child (FCI)	+	2298	0.05 (1.21)	0.00 (1.21)	-0.01	-0.01	[-0.11, 0.09]	0.840	1
2	Caregiver Responsiveness – play responses (<i>total score; range 0-11</i>)	+	2298	7.79 (1.67)	7.81 (1.52)	0.04	0.02	[-0.12, 0.16]	0.731	1
2	Caregiver Responsiveness – non-play responses (<i>total score; range 1-5</i>)	-	2298	1.72 (0.80)	1.79 (0.82)	0.04	0.05	[-0.08, 0.18]	0.527	1
2	Positive Parenting Discipline (MICS)	+	2298 ^e	-0.05 (0.68)	-0.07 (0.68)	-0.02	-0.02	[-0.12, 0.07]	0.620	1
2	Negative Parenting Discipline (MICS)	-	2298	0.02 (0.51)	0.00 (0.50)	-0.01	-0.02	[-0.12, 0.07]	0.626	1
3	Child pro-social behavior (BITSEA) (<i>mean score; range 0-2</i>)	+	2298	1.63 (0.34)	1.64 (0.32)	0.01	0.04	[-0.05, 0.14]	0.392	0.376
3	Child externalizing behavior (BITSEA) (<i>mean score; range 0-2</i>)	-	2298 ^c	0.54 (0.43)	0.50 (0.43)	-0.03	-0.07	[-0.17, 0.02]	0.131	0.306
3	Child internalizing behavior (BITSEA) (<i>mean score; range 0-2</i>)	-	2298	0.48 (0.35)	0.46 (0.33)	-0.02	-0.06	[-0.16, 0.04]	0.240	0.316
3	Child developmental milestones (ASQ-3) Communication (<i>total score; range 0-12</i>)	+	2298	9.65 (2.73)	9.77 (2.65)	0.18	0.06	[-0.03, 0.16]	0.195	0.306
3	Child developmental milestones (ASQ-3) Gross Motor (<i>total score; range 0-12</i>)	+	2298 ^d	10.41 (2.32)	10.55 (2.16)	0.15	0.06	[-0.03, 0.16]	0.189	0.306
3	Child developmental milestones (ASQ-3) Fine Motor (<i>total score; range 0-12</i>)	+	2298 ^d	7.96 (3.29)	8.26 (3.25)	0.29*	0.09*	[-0.01, 0.19]	0.093	0.306
3	Child developmental milestones (ASQ-3) Problem Solving (<i>total score; range 0-12</i>)	+	2298	9.75 (2.38)	9.99 (2.33)	0.31*	0.13*	[0.03, 0.23]	0.014	0.109
4	CHV encourages health activities (<i>yes/no</i>)	null	2298 ^c	0.84 (0.18)	0.83 (0.19)	-0.06	0.94 (odds ratio)	[-1.60, 1.05]	0.680	1
4	Days spent on health activities after call (<i>total score; range 0-75</i>)	null	2298	4.18 (4.35)	4.24 (3.91)	0.16	0.04	[-0.07, 0.15]	0.517	.796
4	Hours/day spent on health activities (<i>categorical 0-4: 10-20 mins; 21-40; 41-60; 1-2 hours</i>)	null	2298 ^c	1.79 (1.00)	1.75 (1.01)	-0.04	-0.04	[-0.15, 0.07]	0.516	.796
4	CHV encourages ECD activities (<i>yes/no</i>)	+	2298 ^c	0.83 (0.18)	0.85 (0.15)	0.12	1.12 (odds ratio)	[-0.91, 2.06]	0.446	.796
4	Days spent on ECD activities after call (<i>total score; range 0-40</i>)	+	2298 ^c	3.86 (4.05)	3.86 (3.58)	0.05	0.01	[-0.08, 0.11]	0.783	1
4	Hours/day spent ECD activities (<i>categorical 0-4</i>)	+	2298	1.79 (1.15)	1.71 (1.08)	-0.07	-0.06	[-0.18, 0.06]	0.342	.796

^a Sharpened False Discovery Rate (FDR) *q*-values, conducted within conceptual groupings (modeling the groupings of a similar prior RCT: Magnuson et al., 2022; Noble et al., 2021)

^b Where not marked, factor scores are used: factor scores have mean of 0 and SD of 1 across the population CFA ran on (i.e. those who participated in endline) with scores imputed for those that did not

^c Model only ran on 99 of 100 imputed data sets due to convergence issues

^d Model only ran on 98 of 100 imputed data sets due to convergence issues

^e Model only ran on 96 of 100 imputed data sets due to convergence issue

and child outcomes and a parenting program previously proven effective in in-person format was delivered successfully with remote coaching on the phone, following video uploads of parent-child interactions (Crawford et al., 2021). The IRC team chose to deliver calls by audio only due to expressed caregiver preferences (related to concerns about data costs); but it may be worth covering additional costs for caregivers in order to include program components such as video and photos.

It may also be that dimensions of parenting culturally specific to the sample (Syrian and Jordanian caregivers) were not sufficiently targeted. An NGO from the region adapted, piloted, and iteratively revised the RUL activities in response to caregiver feedback. Nevertheless, the model fundamentally remains one that was brought in from other contexts and not one that organically grew from the context. In addition, the scripted nature of the calls may have barred CHVs from tailoring the call to the caregivers' needs further contributing to the possible cultural mismatch. Culturally relevant socialization practices might not have been captured in its theory of change. Further research about child-rearing in this context may open new approaches to enhancing children's development and further center the voices of refugees and those affected by displacement in initial phases of program development (Lakkis et al., 2020; Moving Minds Alliance, 2022).

Our finding of a reduction in caregiver depressive symptoms is a rarity within the general early childhood parenting support literature. A recent comprehensive and global meta-analysis of evaluations of parenting interventions for caregivers of 0-3 year old children found no overall average impact of such programs on depressive symptoms (Jeong et al., 2021). When using a cutoff of moderately severe or severe symptomatology (>1.875 mean score on a scale of 0-3 or a total score of 15 across all 8 items), our findings indicate that 10 percent of the control group experienced this level of symptoms (an indication of risk for a depression diagnosis) while only 7 percent of the treatment group did. This effect occurred from a base level of depressive symptoms that was relatively low (in the range of several times in the past month). This level is lower than levels found using the same measure among a sample of Syrian asylum seekers in a refugee camp in Greece (Poole et al., 2018) but higher than those found in a sample of Syrian refugees in Lebanon (Naal et al., 2021). Our levels are in line with protracted rather than acute humanitarian crisis as reflected in the majority of the sample having lived in Jordan for at least several years.

How might this program have reduced depressive symptoms? Some intervention aspects related to well-being occurred in both groups. For example, all CHVs were trained on how and where to refer families showing distress or mental health problems. This said, three components potentially related to well-being only occurred in the RUL group. First, the curriculum for the RUL adaptation was itself supplemented with a module and booklet for facilitators that was focused on caregiver as well as CHV well-being (in response to the COVID-19 pandemic). Second, a check-in on caregiver well-being was explicitly included at the beginning of the script for each call in the treatment group. Third, treatment group CHVs received more training than the control group CHVs overall, including additional training on how to discuss content with caregivers, listen actively, respond to caregiver questions, and build positive rapport. These three components may have led to a stronger emphasis on caregiver well-being. In addition, the program team reported that caregivers and CHVs in the treatment group reported discussions about well-being and family circumstances, such as relationship with one's spouse and economic stress, and that CHVs were responsive to, and built on, such topics when brought up. This finding is worth exploring further in the research and policy spaces to understand further, the ways through which providing ECD support to caregivers over the phone can be done in an effective way that help reduce caregivers' depression.

4.1. Limitations

Several limitations of this study should be noted. First, all research measures were assessed on the phone. As with the intervention, data

collection was audio-only. In person data collection was not feasible during the study period (October 2020 – January 2022), which was characterized by varying levels of COVID-19 restrictions in Jordan. In-person measures, particularly those directly measuring child development, are generally stronger and more predictive than parent-reported, phone-based assessments (Fernald et al., 2017). However, phone-based assessments of depressive symptoms have shown validity in many studies (Wang et al., 2021). Second, dropout rates from the program were substantial (about 25 % for caregivers and a 19 % CHV turnover rate). Although predictors of dropout from the program did not differ across the treatment arms, in both groups caregivers with higher socioeconomic status were more likely to leave the sample. Third, the caregiver survey was quite long (80 minutes at baseline and 45-50 minutes at endline) which may have resulted in participant fatigue. Fourth, although the measures had been validated previously in extensive in-person piloting (Wuermler et al., 2019), were reviewed by the local Arabic-speaking research team (comprised of researchers from both Jordanian and Syrian backgrounds), and showed acceptable psychometric properties, it may still have been the case that some items may have been interpreted differently than intended, particularly when conveyed on the phone. Finally, the sample was recruited by CHVs from their networks making the generalizability of these results is limited.

4.2. Conclusion and Recommendations

Although the switch from in-person to phone in this context was driven by the COVID-19 pandemic, the modality has relevance for other crisis contexts. With increases in climate and conflict-driven crises worldwide, there may continue to be situations in which caregivers of young children are not available for in-person services, but have working cell phones. This evaluation both suggests that higher dosage for ECD content is likely needed to impact parenting and child development and demonstrates that audio-based programming can improve caregiver well-being (i.e., depressive symptoms). Depression is an uncommon outcome in ECD / parenting support programs despite being the sixth leading cause of global disease burden (Vos et al., 2020). Maternal depression has been linked to child well-being, behavior, growth, and development in a wide range of studies, with indication that concerns around this may be even higher in LMICs (Erui et al., 2018). All of this suggests that a critical additional, but often neglected, focus on caregiver well-being may be overdue in ECD programming.

The original RUL curriculum does not focus on caregiver well-being or depression. Indeed, most curricula for caregivers of 0-3 year-old children focus on health and nutrition and do not have strong emphases on caregiver well-being. This adaptation of RUL for audio-only format included explicit check-ins as to general well-being. The training and coaching used for treatment CHVs in this program on how to cover topics related to well-being and family functioning and how to build rapport with caregivers by phone may be useful for phone-based programs in other contexts that also seek to reduce depressive symptoms. Finally, future parenting interventions during crises may also benefit from placing parenting in broader cultural and contextual theories of change that are anchored in caregivers' strengths, needs, values, faith, and coping strategies.

CRedit authorship contribution statement

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Declaration of competing interest

We have no known conflict of interest to disclose.

Data availability

The data that has been used is confidential.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ecresq.2024.07.004.

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